Elementary First-Order Logic

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1 Tethering

A baby instinctively moves to the softer floor to crawl on, directly engaging with the concept of "softer." Similarly, an ignored cry for attention is followed by a "louder" one. This occurs years before the child comprehends the words "carpet," "floorboards," "crying," or "screaming."

The goal of a scientist is to step outside oneself. The scientific method is built upon experiment: to isolate and observe is to analyze beyond bias. A young scientist often begins their journey convinced of the method's success. But after encountering skepticism—perhaps through the study of the foundations of mathematics—they may come to believe that we are bound by our senses, that the scientific method is ultimately constrained by human perception. However, even the most cynical mathematician acknowledges that something *happens* when we perform mathematics (consider, for instance, Wigner's famous essay title, "The Unreasonable Effectiveness of Mathematics in the Natural Sciences"). So what is it?

Perhaps mathematics is entirely within our minds, or perhaps it reflects our cognitive structures. Perhaps it emerges from the game-theoretic strategies that helped our ancestors survive, encoded into our neural architecture. But to what do we apply these strategies? If they correspond to some objective reality independent of us, then we return to our initial belief: that the scientific method works. Yet now, objectivity is not about how the universe *is* but rather about how we *interact* with it. Thus, while the scientific method is inevitably entangled with human perception, we can still be objective about this very entanglement.

If mathematics is psychology, then how do we apply it? Let us not abandon the claim that we *do* perceive. Consider that our interactions with the world are legitimate: I do not know what grass is, but I know that it is softer than concrete (whatever concrete is). I do not know what a mouth is, but I know when its noise is insufficient to gain attention. The *relationships* between things present themselves first—they are undeniable. Relations between what? Some kind of objects, one might suppose. But is this truly so clear? When asked what a horse is, I end up telling a story. This story contains relations, so we are back where we started. The classic response is that "it's there, but we cannot interact with it directly." My thesis, stated vaguely, is this:

why not conflate "out of reach" with "unreal"? Let us be bold: I cannot interact with isolated objects, but I can relate to stories and relationships. Hence, objects *do not exist*, but abstract relations *do*.

2 Syntax: First-Order Languages and Theories

Most introductions to foundations leave implicit what is being asked of the reader (with some exceptions; see [Murf25]). In this section we first present the *standard* (set-theoretic) formation of many-sorted first-order syntax as it appears in classical texts (e.g. [Ende01]). We then explain a common source of dissatisfaction with that presentation, and give an *effective* (computable) reformulation that addresses the relevant concerns while keeping the usual mathematical content.

2.1 Standard formation (set-theoretic presentation)

Definition 2.1. A (many-sorted) **first-order signature** Σ consists of:

- a set Σ-Sort of **sorts**;
- for each $A \in \Sigma\text{-Sort}$, a countably infinite set $\mathcal{V}_A$ of **variables** of sort A;
- a set Σ-Fun of **function symbols**, where each $f \in \Sigma\text{-Fun}$ has a specified **type**

  $$f : A_1 \times \cdots \times A_n \to B \qquad (n \geq 0),$$

  with $A_i, B \in \Sigma\text{-Sort}$;
- a set Σ-Rel of **relation symbols**, where each $R \in \Sigma\text{-Rel}$ has a specified **type**

  $$R \rightarrowtail A_1 \times \cdots \times A_n \qquad (n \geq 0),$$

  with $A_i \in \Sigma\text{-Sort}$.

When $n = 0$, a function symbol is a **constant symbol** and a relation symbol is a **nullary predicate** (an atomic proposition).

Definition 2.2. Fix a signature Σ. For each sort $A \in \Sigma\text{-Sort}$ we define the set $\mathrm{Term}_A(\Sigma)$ of **terms of sort** A as the smallest collection satisfying:

- if $x \in \mathcal{V}_A$, then $x \in \mathrm{Term}_A(\Sigma)$;
- if $f : A_1 \times \cdots \times A_n \to B$ is a function symbol and $t_i \in \mathrm{Term}_{A_i}(\Sigma)$, then

  $$f(t_1, \ldots, t_n) \in \mathrm{Term}_B(\Sigma).$$

Write $\mathrm{Term}(\Sigma) = \bigcup_A \mathrm{Term}_A(\Sigma)$.

Define the set $\mathrm{FV}(t)$ of **free variables of a term** t recursively by

$$\mathrm{FV}(x) = \{x\}, \qquad \mathrm{FV}(f(t_1, \dots, t_n)) = \bigcup_{i=1}^{n} \mathrm{FV}(t_i).$$

Definition 2.3. Fix a signature Σ. We define the set $\mathrm{Form}(\Sigma)$ of **formulas** and their free variables $\mathrm{FV}(\varphi)$ by recursion:

- (Atomic) If $R \rightarrowtail A_1 \times \cdots \times A_n$ and $t_i \in \mathrm{Term}_{A_i}(\Sigma)$, then

$$R(t_1, \dots, t_n) \in \mathrm{Form}(\Sigma), \qquad \mathrm{FV}(R(t_1, \dots, t_n)) = \bigcup_{i=1}^{n} \mathrm{FV}(t_i).$$

- (Equality) If $s, t \in \mathrm{Term}_A(\Sigma)$, then

$$s = t \in \mathrm{Form}(\Sigma), \qquad \mathrm{FV}(s = t) = \mathrm{FV}(s) \cup \mathrm{FV}(t).$$

- $\top, \bot \in \mathrm{Form}(\Sigma)$ with $\mathrm{FV}(\top) = \mathrm{FV}(\bot) = \varnothing$.
- If $\varphi, \psi \in \mathrm{Form}(\Sigma)$ then $\varphi \wedge \psi$, $\varphi \vee \psi$, $\varphi \to \psi \in \mathrm{Form}(\Sigma)$ and

$$\mathrm{FV}(\varphi \circ \psi) = \mathrm{FV}(\varphi) \cup \mathrm{FV}(\psi) \quad (\circ \in \{\wedge, \vee, \to\}).$$

- If $\varphi \in \mathrm{Form}(\Sigma)$ then $\neg\varphi \in \mathrm{Form}(\Sigma)$ and $\mathrm{FV}(\neg\varphi) = \mathrm{FV}(\varphi)$.
- If $x \in \mathcal{V}_A$ and $\varphi \in \mathrm{Form}(\Sigma)$ then $(\exists x : A)\varphi$ and $(\forall x : A)\varphi$ are formulas with

$$\mathrm{FV}((Qx : A)\varphi) = \mathrm{FV}(\varphi) \setminus \{x\} \quad (Q \in \{\exists, \forall\}).$$

Definition 2.4. A **sentence** is a formula $\sigma \in \mathrm{Form}(\Sigma)$ with $\mathrm{FV}(\sigma) = \varnothing$.

A **(first-order) theory** over Σ is a set T of Σ-sentences.

Example 2.5. Let Σ_{grp} have one sort A and function symbols

$$\cdot : A \times A \to A, \qquad (-)^{-1} : A \to A, \qquad e : A.$$

The **theory of groups** T_{grp} is the set of sentences:

$$\begin{aligned}
&(\forall x : A)(\forall y : A)(\forall z : A)\big((x \cdot y) \cdot z = x \cdot (y \cdot z)\big),\\
&(\forall x : A)\big(x \cdot x^{-1} = e\big),\\
&(\forall x : A)\big(e \cdot x = x\big),\\
&(\forall x : A)\big(x \cdot e = x\big).
\end{aligned}$$

Remark 2.6. Even the standard formation above is not philosophically "closed." We have the following complaints:

- In Definition 2.1, what do we mean by a *set*? ZFC provides one account, but ZFC is itself a first-order theory; hence, as a foundational starting-point, the use of the word "set" can feel circular.
- In Definition 2.1, why do we allow *arbitrary* infinite sets of symbols? The definition permits symbol-collections whose membership cannot be determined by any finite means. For example, one can choose a countably infinite set $V \subset \mathbb{R}$ consisting of irrationals such that there is no effective method to decide, given a real r, whether $r \in V$. This can clash with a "finitary access" intuition: in practice we want to be able to identify, distinguish, and generate symbols on demand.

Is there any hope of addressing the concerns raised in Remark 2.6? One approach is simply to accept the standard definitions as definitions of *mathematical objects* in whatever background mathematics we are using: just as one defines groups and vector spaces, one defines signatures, formulas, and theories. The substantive question then becomes whether these objects (and the theorems about them) are of genuine interest.

A different approach is to refine the syntax so that it *builds in* the finitary access that working mathematicians actually use. This does not attempt to eliminate all background commitments, but it makes explicit what it means to "have access" to an infinite reservoir of symbols: namely, that there is an effective procedure for generating and comparing them. This motivates an *effective* (computable) presentation of syntax.

2.2 Effective formation (computable presentation)

The guiding constraint is:

> We only ever handle *finite* expressions, but we require that fresh symbols can be generated and compared whenever needed by an effective procedure.

A Turing machine is a standard formalization of "effective procedure" [BBJ07]. Accordingly, we will treat symbol-collections and syntactic classes as being specified by algorithms operating on finite codes (e.g. Gödel codes) [Ende01].

Definition 2.7. A subset $X \subseteq \mathbb{N}$ is **decidable** if there is a Turing machine which, on input n, halts and answers whether $n \in X$. A subset $X \subseteq \mathbb{N}$ is **computably enumerable** if there is a Turing machine which enumerates exactly the elements of X.

Definition 2.8. An **effective many-sorted first-order signature** is data

$$\Sigma = (\Sigma\text{-Sort}, \Sigma\text{-Fun}, \Sigma\text{-Rel}, \mathrm{type}_{\mathrm{Fun}}, \mathrm{type}_{\mathrm{Rel}})$$

such that:

1. Σ-Sort $\subseteq \mathbb{N}$ is decidable;
2. Σ-Fun $\subseteq \mathbb{N}$ is decidable, and $\mathrm{type}_{\mathrm{Fun}}$ is a computable total function

$$\mathrm{type}_{\mathrm{Fun}} : \Sigma\text{-Fun} \to \bigcup_{n\in\mathbb{N}} \big((\Sigma\text{-Sort})^n \times \Sigma\text{-Sort}\big),$$

so that $\mathrm{type}_{\mathrm{Fun}}(f) = (A_1, \dots, A_n; B)$ is read as $f : A_1 \times \cdots \times A_n \to B$;
3. Σ-Rel $\subseteq \mathbb{N}$ is decidable, and $\mathrm{type}_{\mathrm{Rel}}$ is a computable total function

$$\mathrm{type}_{\mathrm{Rel}} : \Sigma\text{-Rel} \to \bigcup_{n\in\mathbb{N}} (\Sigma\text{-Sort})^n,$$

so that $\mathrm{type}_{\mathrm{Rel}}(R) = (A_1, \dots, A_n)$ is read as $R \rightarrowtail A_1 \times \cdots \times A_n$.

Definition 2.9. Given an effective signature Σ, define the set of **variables** (as codes) by

$$\Sigma\text{-Var} := \Sigma\text{-Sort} \times \mathbb{N}.$$

We think of (A, i) as "the i-th variable of sort A."

If $V \subseteq \Sigma$-Var is finite, define the **fresh-variable operator**

$$\mathrm{fresh}_A(V) := (A, i)$$

where i is the least natural number such that $(A, i) \notin V$.

Remark 2.10. The standard formation postulates countably infinite variable-reservoirs $\mathcal{V}_A$. The effective formation replaces this with an explicit *procedure*: given finitely many variables already in use, one can compute a fresh one. This is exactly what "finitary access" demands.

Next, one fixes a concrete coding of finite trees (terms and formulas) by natural numbers. Any standard primitive-recursive Gödel coding suffices [Ende01]. With such a coding in place, we can regard "being a well-formed term/formula" as a predicate on $\mathbb{N}$.

Definition 2.11. Fix an effective signature Σ and a Gödel coding of finite syntax trees by $\mathbb{N}$.

- For each sort $A \in \Sigma$-Sort, let $\mathrm{Term}_A^{\mathrm{eff}}(\Sigma) \subseteq \mathbb{N}$ be the set of codes of well-formed Σ-terms of sort A.
- Let $\mathrm{Form}^{\mathrm{eff}}(\Sigma) \subseteq \mathbb{N}$ be the set of codes of well-formed Σ-formulas.

These sets are defined by structural recursion on codes using the computable type-functions $\mathrm{type}_{\mathrm{Fun}}$ and $\mathrm{type}_{\mathrm{Rel}}$; in particular, membership in $\mathrm{Term}_A^{\mathrm{eff}}(\Sigma)$ and $\mathrm{Form}^{\mathrm{eff}}(\Sigma)$ is decidable.

Definition 2.12. An **effectively axiomatized theory** over an effective signature Σ is a computably enumerable set $T \subseteq \mathbb{N}$ such that every element of T codes a Σ-sentence. Equivalently, there is a Turing machine which enumerates exactly the Gödel codes of the axioms of T.

Remark 2.13. The effective formation addresses the dissatisfactions in Remark 2.6 as follows.

- It does not attempt to settle the philosophical question "what is a set?" but it *replaces arbitrary symbol-sets* with decidable predicates (or equivalently Turing machines) on $\mathbb{N}$. Thus the assumed background is closer to computation on finite data than to unrestricted set theory.
- It prevents pathologies of "symbol collections with no finite recognizability" by requiring decidable membership and computable typing data. Infinite reservoirs persist, but only as *computably presented* infinities: one can generate new symbols and verify their types effectively.

In particular, the effective presentation is a principled way to formalize the operational content of the standard formation: it makes explicit what it means to have access to infinitely many symbols while only ever manipulating finite expressions.

3 Semantics: Structures and Satisfaction

Up to this point we have treated *syntax* (signatures, terms, formulas) and *proof* as formal objects. To speak about *models*, *truth*, and *satisfaction*, we must additionally be explicit about the *metatheory* in which these semantic notions are defined.

Informally, a *metatheory* is the background mathematical system in which we are allowed to say things like:

> "$\mathcal{M}(A)$ is a set," "$f^{\mathcal{M}}$ is a function," "$\exists a \in \mathcal{M}(A)$," and "define $(\mathcal{M}, s) \models \varphi$ by recursion on the shape of φ."

Without choosing a metatheory, statements such as "$\mathcal{M} \models T$" are not yet well-formed: the symbol $\models$ is not part of the object-language Σ, but rather a *metatheoretic predicate* that we define *about* Σ-formulas.

Most texts take the metatheory to be classical set theory (classical logic + ZFC) and proceed silently. Here we choose a metatheory compatible with the constructive stance developed earlier.

The chosen metatheory

Remark 3.1. In this section, we fix the metatheory to be **constructive first-order logic** together with the usual ZF-style axioms for sets, i.e. *constructive ZFC* (often called *IZF* in the literature). Thus:

- all metatheoretic connectives $\wedge, \vee, \to, \neg$ and quantifiers $\forall, \exists$ are interpreted *constructively*;
- semantic definitions may quantify over sets and functions, but only using the constructive logic rules available in the metatheory.

This is a choice of *ambient mathematics*, not an additional axiom of the object-theories we will study. (For classical presentations of Tarski semantics one may compare [Ende01].)

Σ-structures

Definition 3.2. Σ-structures. Let Σ be a (many-sorted) first-order signature. A **Σ-structure** (also called a **model** or **interpretation**) $\mathcal{M}$ consists of:

- for each sort $A \in \Sigma\text{-Sort}$, an *inhabited* set $\mathcal{M}(A)$ (i.e. $\exists a \in \mathcal{M}(A)$);
- for each function symbol $f : A_1 \times \cdots \times A_n \to B$, a function
$$f^{\mathcal{M}} : \mathcal{M}(A_1) \times \cdots \times \mathcal{M}(A_n) \to \mathcal{M}(B);$$
- for each relation symbol $R \rightarrowtail A_1 \times \cdots \times A_n$, a subset
$$R^{\mathcal{M}} \subseteq \mathcal{M}(A_1) \times \cdots \times \mathcal{M}(A_n).$$

Remark 3.3. In classical metatheory, one usually requires each carrier $\mathcal{M}(A)$ to be *nonempty*. In constructive metatheory, it is often cleaner to require *inhabitedness* (an explicit existential witness), because "$\mathcal{M}(A) \neq \varnothing$" does not in general provide an element without additional principles. For ordinary mathematics over inhabited carriers, the two notions behave as expected.

Valuations without hidden choice

A common (classical) definition of an assignment is a family of total functions $s_A : \mathcal{V}_A \to \mathcal{M}(A)$ over all variable-reservoirs. Constructively, the existence of such global families can hide choice principles when there are many sorts/variables. Since every term or formula mentions only finitely many variables, it is more faithful to mathematical practice to evaluate relative to valuations on a *finite* variable-set.

Definition 3.4. Finite valuations. Let $\mathcal{M}$ be a Σ-structure. Let V be a finite set of variables, where each variable comes with a declared sort. A **valuation** (assignment) on V in $\mathcal{M}$ is a family of functions

$$s_A : \big(V \cap \mathcal{V}_A\big) \to \mathcal{M}(A) \qquad (A \in \Sigma\text{-Sort}).$$

If $x \in \mathcal{V}_A$ and $a \in \mathcal{M}(A)$, define the updated valuation $s[x \mapsto a]$ on $V \cup \{x\}$ by sending x to a and agreeing with s on all other variables.

Remark 3.5. If σ is a *sentence*, then $\mathrm{FV}(\sigma) = \varnothing$, so σ is evaluated using the unique valuation on the empty set. This avoids any need to choose a global assignment "for all variables" in the metatheory.

Interpretation of terms and satisfaction

Definition 3.6. Interpretation of terms and satisfaction (constructive metatheory). Let $\mathcal{M}$ be a Σ-structure. Let t be a term and φ a formula. Let V be a finite set of variables containing all variables occurring in t or φ (e.g. $V = \mathrm{FV}(t)$ or $V = \mathrm{FV}(\varphi)$), and let s be a valuation on V in $\mathcal{M}$.

Interpretation of terms. Define $[\![t]\!]_{\mathcal{M},s}$ by recursion:

$$[\![x]\!]_{\mathcal{M},s} = s(x), \qquad [\![f(t_1, \dots, t_n)]\!]_{\mathcal{M},s} = f^{\mathcal{M}}([\![t_1]\!]_{\mathcal{M},s}, \dots, [\![t_n]\!]_{\mathcal{M},s}).$$

Satisfaction. Define the metatheoretic proposition $(\mathcal{M}, s) \models \varphi$ by recursion on φ:

- (Atomic relations) $(\mathcal{M}, s) \models R(t_1, \dots, t_n)$ iff
$$([\![t_1]\!]_{\mathcal{M},s}, \dots, [\![t_n]\!]_{\mathcal{M},s}) \in R^{\mathcal{M}}.$$
- (Equality) $(\mathcal{M}, s) \models (t_1 = t_2)$ iff
$$[\![t_1]\!]_{\mathcal{M},s} = [\![t_2]\!]_{\mathcal{M},s}.$$
- $(\mathcal{M}, s) \models \top$ always, and $(\mathcal{M}, s) \models \bot$ never.
- (Connectives) Using the connectives of the *constructive metatheory*,
$$(\mathcal{M}, s) \models (\varphi \wedge \psi) \iff (\mathcal{M}, s) \models \varphi \ \wedge \ (\mathcal{M}, s) \models \psi,$$
$$(\mathcal{M}, s) \models (\varphi \vee \psi) \iff (\mathcal{M}, s) \models \varphi \ \vee \ (\mathcal{M}, s) \models \psi,$$
$$(\mathcal{M}, s) \models (\varphi \to \psi) \iff \big((\mathcal{M}, s) \models \varphi\big) \ \to \ \big((\mathcal{M}, s) \models \psi\big),$$
and $(\mathcal{M}, s) \models \neg\varphi$ means $(\mathcal{M}, s) \models \varphi \to \bot$.
- (Quantifiers) If $x \in \mathcal{V}_A$, then
$$(\mathcal{M}, s) \models (\exists x : A)\, \varphi \iff \exists a \in \mathcal{M}(A) : \ (\mathcal{M}, s[x \mapsto a]) \models \varphi,$$
$$(\mathcal{M}, s) \models (\forall x : A)\, \varphi \iff \forall a \in \mathcal{M}(A) : \ (\mathcal{M}, s[x \mapsto a]) \models \varphi.$$

Models of sentences and theories

Definition 3.7. Models, satisfiability, entailment. Let σ be a Σ-sentence and let T be a Σ-theory (a set of Σ-sentences).

- Define $\mathcal{M} \models \sigma$ to mean $(\mathcal{M}, \emptyset) \models \sigma$, where $\emptyset$ is the unique valuation on $\varnothing$.
- Define $\mathcal{M} \models T$ to mean $\forall \sigma \in T,\ \mathcal{M} \models \sigma$.

- Define **satisfiability** of T by the metatheoretic statement

$$\mathrm{Sat}(T) \;:\Longleftrightarrow\; \exists \mathcal{M}\ \big(\mathcal{M} \text{ is a } \Sigma\text{-structure and } \mathcal{M} \models T\big).$$

- Define semantic entailment by

$$T \models \sigma \;:\Longleftrightarrow\; \forall \mathcal{M}\ \big(\mathcal{M} \models T \;\to\; \mathcal{M} \models \sigma\big).$$

Remark 3.8. It is crucial not to conflate *satisfaction* with *provability in the metatheory*. The statement $\mathcal{M} \models \sigma$ is (by Definition 3.6) a metatheoretic proposition about certain sets, functions, and membership/equality facts *in the ambient universe*. If we additionally want to assert that satisfiability is *provable* in the ambient system, we write

$$\mathrm{IZF} \vdash \mathrm{Sat}(T),$$

which is stronger than merely $\mathrm{Sat}(T)$.

Remark 3.9. Everything above is phrased in a set-theoretic metatheory (constructive ZF-style sets). Later, one can replace this ambient semantics by categorical semantics: for example, interpreting first-order logic internally in a topos [MacM94, John02]. In that setting, "$\models$" is again a metatheoretic notion, but the ambient universe is no longer **Set**.

Models and satisfiability

Definition 3.10. Let σ be a Σ-sentence and T a Σ-theory (a set of Σ-sentences).

- Define $\mathcal{M} \models \sigma$ to mean $(\mathcal{M}, \emptyset) \models \sigma$, where $\emptyset$ is the unique valuation on $\varnothing$.
- Define $\mathcal{M} \models T$ to mean $\forall \sigma \in T,\ \mathcal{M} \models \sigma$.
- Define **satisfiability** of T by

$$\mathrm{Sat}(T) \;:\Longleftrightarrow\; \exists \mathcal{M}\ \big(\mathcal{M} \text{ is a } \Sigma\text{-structure and } \mathcal{M} \models T\big).$$

- Define semantic entailment by

$$T \models \sigma \;:\Longleftrightarrow\; \forall \mathcal{M}\big(\mathcal{M} \models T \;\to\; \mathcal{M} \models \sigma\big).$$

Remark 3.11. The statement $\mathrm{Sat}(T)$ is a *metatheoretic* existence claim. If one wants to assert that satisfiability is *provable* in the ambient theory (IZF), one writes

$$\mathrm{IZF} \vdash \mathrm{Sat}(T),$$

which is stronger than $\mathrm{Sat}(T)$ itself.

4 Proof: Natural Deduction

We present a standard classical natural deduction system for first-order logic with equality (cf. [Dale13]).

Definition 4.1. Natural deduction rules. We use the usual introduction/elimination rules for $\wedge, \vee, \rightarrow, \neg, \forall, \exists$, plus equality and $\bot$.

Conjunction

$$\frac{\varphi \qquad \psi}{\varphi \wedge \psi}\,\wedge I \qquad \frac{\varphi \wedge \psi}{\varphi}\,\wedge E_1 \qquad \frac{\varphi \wedge \psi}{\psi}\,\wedge E_2$$

Disjunction

$$\frac{\varphi}{\varphi \vee \psi}\,\vee I_1 \qquad \frac{\psi}{\varphi \vee \psi}\,\vee I_2 \qquad \frac{\varphi \vee \psi \qquad \begin{array}{c}[\varphi]^i \\ \vdots \\ \delta\end{array} \qquad \begin{array}{c}[\psi]^j \\ \vdots \\ \delta\end{array}}{\delta}\,\vee E^{i,j}$$

Implication

$$\frac{\begin{array}{c}[\varphi]^i \\ \vdots \\ \psi\end{array}}{\varphi \rightarrow \psi}\rightarrow I^i \qquad \frac{\varphi \rightarrow \psi \qquad \varphi}{\psi}\rightarrow E$$

Negation and classical reasoning

$$\frac{\begin{array}{c}[\varphi]^i \\ \vdots \\ \bot\end{array}}{\neg\varphi}\,\neg I^i \qquad \frac{\neg\varphi \qquad \varphi}{\bot}\,\neg E \qquad \frac{\begin{array}{c}[\neg\varphi]^i \\ \vdots \\ \bot\end{array}}{\varphi}\,RAA^i$$

Quantifiers (with standard freshness conditions)

$$\frac{\varphi[x := y]}{(\forall x : A)\varphi}\,\forall I \qquad \frac{(\forall x : A)\varphi}{\varphi[x := t]}\,\forall E$$

The $\forall I$ rule requires that y is not free in any undischarged assumption on which $\varphi[x := y]$ depends.

$$\frac{\varphi[x := t]}{(\exists x : A)\varphi}\,\exists I \qquad \frac{(\exists x : A)\varphi \qquad \begin{array}{c}[\varphi[x := y]]^i \\ \vdots \\ \gamma\end{array}}{\gamma}\,\exists E^i$$

The $\exists E$ rule requires that y is not free in γ nor in any undischarged assumption used in the derivation of γ (except the temporary assumption $\varphi[x := y]$).

Falsum and equality

$$\frac{\bot}{\varphi}\ \bot E \qquad\qquad \frac{s = t \qquad \varphi[x := s]}{\varphi[x := t]} = E$$

A **derivation** (proof) is a finite rooted tree labeled by instances of these rules, with leaves labeled by assumptions. If there is a derivation of φ from assumptions in Γ, we write $\Gamma \vdash \varphi$.

Definition 4.2. Consistency. A set of sentences T is **(syntactically) consistent** if $T \nvdash \bot$.

Lemma 4.3. *Finite character of proofs.* *If $T \vdash \varphi$, then there exists a finite $T_0 \subseteq T$ such that $T_0 \vdash \varphi$.*

Proof. Any derivation is a finite tree and therefore uses only finitely many assumptions from T. □

Truth and proof.

How does proof relate to truth? At minimum, they live on different sides of the syntax/semantics divide.

A *proof* is a finite combinatorial object: a derivation in a fixed deductive system. In particular, once the inference rules are specified, "$T \vdash \sigma$" is (at least in principle) a claim about the existence of a finite certificate that can be mechanically checked.

A *truth-claim*, by contrast, is semantic: it is a claim about what holds in an interpretation. Given a structure (model) $\mathcal{M}$ for a signature Σ, we write $\mathcal{M} \models \sigma$ to mean that σ holds in $\mathcal{M}$ in the sense of Tarski semantics. For a theory T, the semantic consequence relation

$$T \models \sigma$$

means: every Σ-structure satisfying T also satisfies σ.

Soundness and completeness connect these notions: soundness says that proofs do not create falsehoods (if $T \vdash \sigma$ then $T \models \sigma$), while completeness says that first-order proof rules are adequate for semantic consequence (if $T \models \sigma$ then $T \vdash \sigma$) [Ende01].

This still leaves a philosophical question: *which* semantics is the intended one? For example, we are confident that $1 \neq 0$ in the intended integers, but a formal theory is only as good as the interpretation we intend it to describe. If a foundational theory entailed $1 = 0$, we would not conclude that arithmetic has changed; rather, we would conclude that something has gone wrong in our formalization (indeed, in classical logic an inconsistency forces every sentence to be provable).

One way to make the dependence on semantics explicit is to allow interpretations in different "universes of discourse." Besides the usual universe of sets, one can interpret first-order (and often richer) syntax in the internal logic of a topos, where the ambient logic may be intuitionistic [MacM94, John02]. On this view, truth is not a free-floating property of strings of symbols: a sentence is true *relative to an interpretation*, i.e. relative to a chosen semantic world and a chosen model inside it.

5 Isomorphisms preserve truth

Definition 5.1. Isomorphism of Σ-structures. Let $\mathcal{M}, \mathcal{N}$ be Σ-structures. An **isomorphism** $\eta : \mathcal{M} \to \mathcal{N}$ is a family of bijections

$$\eta_A : \mathcal{M}(A) \to \mathcal{N}(A) \qquad (A \in \Sigma\text{-Sort})$$

such that:

- for each function symbol $f : A_1 \times \cdots \times A_n \to B$,

$$\eta_B\big(f^{\mathcal{M}}(a_1, \ldots, a_n)\big) = f^{\mathcal{N}}\big(\eta_{A_1}(a_1), \ldots, \eta_{A_n}(a_n)\big);$$

- for each relation symbol $R \rightarrowtail A_1 \times \cdots \times A_n$,

$$(a_1, \ldots, a_n) \in R^{\mathcal{M}} \iff (\eta_{A_1}(a_1), \ldots, \eta_{A_n}(a_n)) \in R^{\mathcal{N}}.$$

Lemma 5.2. ***Transport of term-values.*** *Let $\eta : \mathcal{M} \to \mathcal{N}$ be an isomorphism and let s be an assignment in $\mathcal{M}$. Define the transported assignment $\eta \circ s$ in $\mathcal{N}$ by $(\eta \circ s)(x) = \eta(s(x))$. Then for every term t,*

$$\eta\big([\![t]\!]_{\mathcal{M},s}\big) = [\![t]\!]_{\mathcal{N},\eta\circ s}.$$

Proof. By induction on the construction of t. If $t = x$, this is immediate from the definition of $\eta \circ s$. If $t = f(t_1, \ldots, t_n)$, apply the induction hypothesis to each t_i and then use preservation of function symbols by η. □

Lemma 5.3. ***Isomorphism invariance.*** *Let $\eta : \mathcal{M} \to \mathcal{N}$ be an isomorphism. For every formula φ and every assignment s in $\mathcal{M}$,*

$$(\mathcal{M}, s) \models \varphi \iff (\mathcal{N}, \eta \circ s) \models \varphi.$$

In particular, for every sentence σ,

$$\mathcal{M} \models \sigma \iff \mathcal{N} \models \sigma.$$

Proof. Induct on the construction of φ.

Atomic cases. For equality $t_1 = t_2$ use Lemma 5.2 and bijectivity of η. For $R(t_1, \ldots, t_n)$ use Lemma 5.2 to transport the tuple of term-values and then use the defining condition for relation symbols in an isomorphism.

Boolean connectives. Immediate from the inductive hypothesis and classical truth conditions.

Quantifiers. For $\exists x : A\, \psi$, we have:

$$(\mathcal{M}, s) \models \exists x\, \psi \iff \exists a \in \mathcal{M}(A) : (\mathcal{M}, s[x \mapsto a]) \models \psi.$$

Apply the induction hypothesis to ψ, and use bijectivity of η_A to rewrite witnesses a in $\mathcal{M}(A)$ as witnesses $\eta_A(a)$ in $\mathcal{N}(A)$. The $\forall$ case is similar. □

6 Soundness, Completeness, and Compactness

We state three cornerstone metatheorems of first-order logic; standard proofs can be found in classical references such as [Ende01].

Theorem 6.1. ***Soundness.*** *If $T \vdash \sigma$ (where T is a set of sentences and σ is a sentence), then $T \models \sigma$.*

Proof. Proof sketch. Induct on the given derivation and verify each rule preserves semantic truth under the Tarski clauses in Definition 3.6. □

Theorem 6.2. ***Gödel Completeness Theorem.*** *If $T \models \sigma$, then $T \vdash \sigma$. Equivalently: if T is consistent, then T has a model.*

Proof. Henkin construction (structured sketch). Assume T is consistent. We outline the standard Henkin proof that T has a model.

(1) Henkin expansion. Extend the language by adding a new constant $c_{\exists x\,\varphi}$ for each formula of the form $\exists x\,\varphi(x)$ (in the expanded language as the construction proceeds), and add the corresponding Henkin axiom

$$(\exists x\,\varphi(x)) \to \varphi(c_{\exists x\,\varphi}).$$

Let T_H be T together with all Henkin axioms.

(2) Lindenbaum extension to a complete theory. Using a standard Lindenbaum argument (enumerate sentences and add each or its negation when this preserves consistency), extend T_H to a *maximal consistent* (hence complete) set of sentences Δ.

(3) Term model. Let D be the set of closed terms modulo provable equality:

$$[t] = \{u \mid \Delta \vdash t = u\}.$$

Interpret function symbols by

$$f^{\mathcal{M}}([t_1], \ldots, [t_n]) = [f(t_1, \ldots, t_n)],$$

and interpret relation symbols by

$$([t_1], \ldots, [t_n]) \in R^{\mathcal{M}} \iff \Delta \vdash R(t_1, \ldots, t_n).$$

Well-definedness uses congruence properties of $=$ derivable in natural deduction.

(4) Truth lemma. Show by induction on formulas that for every *sentence* σ,

$$\mathcal{M} \models \sigma \iff \sigma \in \Delta.$$

The existential step is where Henkin axioms are used to guarantee witnesses among closed terms.

(5) Conclusion. Since $T \subseteq \Delta$, we have $\mathcal{M} \models T$. Thus every consistent theory has a model, proving completeness. □

Theorem 6.3. ***Compactness Theorem.*** *Let T be a first-order theory (a set of sentences). Then T has a model if and only if every finite $T_0 \subseteq T$ has a model.*

Proof. The forward implication is immediate.

For the converse, assume every finite $T_0 \subseteq T$ has a model. If T were inconsistent, then $T \vdash \perp$. By Lemma 4.3 there is a finite $T_0 \subseteq T$ with $T_0 \vdash \perp$. By soundness (Theorem 6.1), T_0 would then have no model, contradicting the hypothesis. Hence T is consistent, so by completeness (Theorem 6.2) it has a model. □

Definition 6.4. Complete theory. A theory T is **complete** if for every sentence σ,

$$T \vdash \sigma \quad \text{or} \quad T \vdash \neg\sigma.$$

7 Skolemization and Löwenheim–Skolem

The following constructions and cardinality theorems are standard in model theory; see [Hodg97].

Definition 7.1. Skolemization. Let T be a theory in a language Σ. A **Skolemization** of T is a theory $\mathcal{S}(T)$ in an expanded language Σ_S obtained by (i) putting axioms of T into prenex form and (ii) replacing each existential quantifier $\exists y$ with a fresh function symbol whose arguments are the preceding universal variables, together with axioms asserting that the function witnesses the existential.

Lemma 7.2. ***Skolemization preserves satisfiability.*** *T has a model if and only if $\mathcal{S}(T)$ has a model. Moreover, if $\mathcal{N} \models \mathcal{S}(T)$ then the reduct of $\mathcal{N}$ to the original language Σ is a model of T.*

Proof. ($\Rightarrow$) Let $\mathcal{M} \models T$. For each Skolem axiom produced from a sentence

$$\forall \bar{x}\, \exists y\, \psi(\bar{x}, y)$$

choose (using Choice) a function $F : \mathcal{M}(\bar{x}) \to \mathcal{M}(y)$ sending each $\bar{a}$ to a witness b with $\mathcal{M} \models \psi(\bar{a}, b)$. Interpret the corresponding Skolem function symbol by F. Doing this for all Skolem symbols expands $\mathcal{M}$ to a Σ_S-structure satisfying all Skolem axioms, hence $\mathcal{S}(T)$.

($\Leftarrow$) If $\mathcal{N} \models \mathcal{S}(T)$ then each Skolem axiom

$$\forall \bar{x}\, \psi(\bar{x}, f(\bar{x}))$$

implies the original sentence $\forall \bar{x}\, \exists y\, \psi(\bar{x}, y)$ in the reduct. Thus the reduct $\mathcal{N}|_\Sigma$ satisfies all axioms of T. □

Lemma 7.3. ***Downward Löwenheim–Skolem (countable language version).*** *Assume Σ is countable and T is a Σ-theory with an infinite model. Then for every infinite cardinal κ with $\aleph_0 \leq \kappa \leq |\mathcal{M}|$ (for some model $\mathcal{M} \models T$), there exists $\mathcal{N} \models T$ with $|\mathcal{N}| = \kappa$.*

Proof. Let $\mathcal{M} \models T$ be infinite and fix $D \subseteq |\mathcal{M}|$ with $|D| = \kappa$. Let Σ_S be a Skolem expansion of Σ for T; since Σ is countable, so is Σ_S. By Lemma 7.2, expand $\mathcal{M}$ to a Σ_S-structure $\mathcal{M}_S \models \mathcal{S}(T)$.

Let $E_0 = D$ and define E_{n+1} by closing E_n under all function symbols of Σ_S:

$$E_{n+1} = E_n \ \cup \ \{ f^{\mathcal{M}_S}(\bar{e}) \mid f \in \Sigma_S\text{-Fun}, \ \bar{e} \in E_n^{\mathrm{arity}(f)} \}.$$

Let $E = \bigcup_{n \in \mathbb{N}} E_n$ and let $\mathcal{N}_S$ be the Σ_S-substructure of $\mathcal{M}_S$ with underlying set E. Because Σ_S is countable and κ is infinite, cardinal arithmetic gives $|E| = \kappa$.

Skolemized theories are universal, hence closed under substructures; therefore $\mathcal{N}_S \models \mathcal{S}(T)$. By Lemma 7.2, the reduct $\mathcal{N} = \mathcal{N}_S|_\Sigma$ satisfies T and has size κ. $\square$

Lemma 7.4. ***Upward Löwenheim–Skolem (countable language version).*** *Assume Σ is countable and T has an infinite model. Then for every infinite cardinal $\kappa \geq \aleph_0$ there exists $\mathcal{N} \models T$ with $|\mathcal{N}| = \kappa$.*

Proof. Expand Σ by adding constants $\{c_\alpha\}_{\alpha<\kappa}$ and let

$$X = \{ c_\alpha \neq c_\beta \mid \alpha < \beta < \kappa \}.$$

Every finite subset of $T \cup X$ has a model: interpret the finitely many constants as distinct elements inside some infinite model of T. By compactness (Theorem 6.3), $T \cup X$ has a model $\mathcal{K}$; in particular $|\mathcal{K}| \geq \kappa$.

Now apply Lemma 7.3 (to the expanded countable language is not possible when κ is uncountable), so instead take $D = \{c_\alpha^{\mathcal{K}} \mid \alpha < \kappa\} \subseteq |\mathcal{K}|$, and perform the same Skolem-hull closure argument *inside* $\mathcal{K}$ using a Skolemization of T in the original countable language. The closure contains D (size κ) and is generated using only countably many function symbols, hence has size exactly κ. Its reduct to Σ is a model of T of size κ. $\square$

Corollary 7.5. ***Löwenheim–Skolem (countable language).*** *If T is a theory in a countable language and T has an infinite model, then T has models of every infinite cardinality.*

Proof. Combine Lemmas 7.3 and 7.4. $\square$

8 A completeness test: the Los–Vaught criterion

The following criterion is a standard tool in model theory; see [Hodg97].

Lemma 8.1. ***Los–Vaught test.*** *Let T be a theory in a countable language such that:*

1. *T has no finite models, and*

2. *T is κ-categorical for some infinite cardinal κ (i.e. any two models of size κ are isomorphic).*

Then T is complete.

Proof. Suppose T is not complete. Then there is a sentence σ such that $T \nvdash \sigma$ and $T \nvdash \neg\sigma$. By completeness (Theorem 6.2), both $T \cup \{\sigma\}$ and $T \cup \{\neg\sigma\}$ have models, say $\mathcal{M} \models T \cup \{\sigma\}$ and $\mathcal{N} \models T \cup \{\neg\sigma\}$.

By assumption (1), both $\mathcal{M}$ and $\mathcal{N}$ are infinite. By the Löwenheim–Skolem corollary (Corollary 7.5), there exist models $\mathcal{M}', \mathcal{N}'$ of sizes κ with

$$\mathcal{M}' \models T \cup \{\sigma\}, \qquad \mathcal{N}' \models T \cup \{\neg\sigma\}.$$

By κ-categoricity, $\mathcal{M}' \cong \mathcal{N}'$. But isomorphic structures satisfy the same sentences (Lemma 5.3), contradiction. Hence T is complete. □

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